

Solution for CS5275 Problem Set 4

Prepared by TA Ivan Adrian Koswara

Contents

1 Problem 1	2
1.1 My solution	2
1.1.1 n queries are sufficient	2
1.1.2 n queries are necessary	2
1.2 Insight and discussion	2
1.3 Grading criteria and common mistakes	3
1.3.1 (Necessity) Major error: Assuming queries are in some form	3
1.4 Appendix: Proof for necessity, based on entropy	3
2 Problem 2	4
2.1 My solution	4
2.2 Insight and discussion	5
2.3 Grading criteria and common mistakes	5
2.4 Appendix: (a) Breaking down the expectation	5
3 Problem 3	7
3.1 My solution	7
3.2 Insight and discussion	7
3.3 Grading criteria and common mistakes	8
4 Problem 4	9
4.1 My solution	9
4.1.1 The conductance is $\leq 1/d$	9
4.1.2 The conductance is $\geq 1/d$	9
4.2 Insight and discussion	10
4.3 Grading criteria and common mistakes	10
4.4 Appendix: An inductive proof without using Boolean functions	10

1 Problem 1

Let $f : \{-1, +1\}^n \rightarrow \{-1, +1\}$ be a parity function. You don't know f , but you may query f at inputs of your choice. Design a deterministic algorithm that determines f using as few queries as possible in the worst case.

1.1 My solution

We first translate the setting into $f : \mathbb{F}_2^n \rightarrow \mathbb{F}_2$, by mapping $+1 \in \{+1, -1\}$ into $0 \in \mathbb{F}_2$, and -1 into 1 . Then because f is a parity function, it is in the form

$$f(\mathbf{x}) = \sum_{i \in S} x_i = \mathbf{1}_S \cdot \mathbf{x}$$

for some unknown set $S \subseteq [n]$, where the sum is taken in $\mathbb{F}_2$. (The main reason for translating the setting is so that we can use linear algebra on $\mathbb{F}_2$, specifically to notice that f is just an inner product.)

We claim that n queries are necessary and sufficient.

1.1.1 n queries are sufficient

We query $f(\mathbf{e}_i)$ for each $i \in [n]$, where $\mathbf{e}_i$ is the unit vector in the i -th direction: it has 1 on the i -th component and 0 on the others. Then

$$f(\mathbf{e}_i) = \begin{cases} 1 & \text{if } i \in S \\ 0 & \text{if } i \notin S \end{cases}$$

Therefore we can just read off which values of f are equal to 1, and those corresponding coordinates exactly make up S .

1.1.2 n queries are necessary

Suppose we query f on $n - 1$ vectors, namely $\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_{n-1}$. Then there exists some nonzero vector $\mathbf{v} \notin \text{span}\{\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_{n-1}\}$. Now consider the following function:

$$f'(\mathbf{x}) = (\mathbf{1}_S + \mathbf{v}) \cdot \mathbf{x}$$

Now, for each i , we have

$$f'(\mathbf{z}_i) = (\mathbf{1}_S + \mathbf{v}) \cdot \mathbf{z}_i = f(\mathbf{z}_i) + \mathbf{0} = f(\mathbf{z}_i)$$

In other words, f' gives the exact same outputs to $\mathbf{z}_i$'s as f . So a deterministic algorithm cannot distinguish these functions, and will be wrong in at least one case.

Therefore, $n - 1$ queries are not enough for any deterministic algorithm, so a deterministic algorithm needs at least n queries.

1.2 Insight and discussion

Strictly speaking, the problem has three parts to it: determine the number of queries (n), prove it is sufficient, and prove it is necessary. I think most of you get the algorithm easily though, which gives both the number of queries and that it is sufficient, so you just need the necessity proof.

One goal of this problem is to ask you to be comfortable to work with Boolean functions in any domain, whether in $\{+1, -1\}^n$ or in $\mathbb{F}_2^n$. Furthermore, it wants to show that parity functions are especially easy to work with. Usually the output of the function should be left in $\{+1, -1\}$, but for parity functions, you don't have to. Treating the domain as $\mathbb{F}_2$ means parity functions are just inner products, and that's so easy to work with. (A general Boolean function will be a sum of multilinear terms where each term is a product of distinct x_i 's, but in a parity function, each term is just a single x_i .)

This is also why my official solution above uses linear algebra on $\mathbb{F}_2$. I understand a vast majority of you used an information theoretic approach, described in Section 1.4, and that is fine. But a linear algebra approach links with the previous course material about spectral graph theory. It finds a vector not in the span of the query vectors, and shows that the function is *never* fully determined with just $n - 1$ queries. (The information theoretic approach does not easily prove this.)

1.3 Grading criteria and common mistakes

1.3.1 (Necessity) Major error: Assuming queries are in some form

When you want to prove the algorithm is optimal, you need to prove that *no* algorithm can achieve better, no matter what queries it makes.

Some of you assume the queries are in a certain form. For example, all its $\mathbf{x}$ will have -1 on exactly one coordinate and $+1$ on everything else. Then yes, it's clear you won't be able to do fewer than n queries: one of the coordinates is never tested. But the algorithm doesn't need all queries to be in this form, and you haven't proven other such algorithms don't work.

Making this error is a major flaw, so it docks around 2 marks.

1.4 Appendix: Proof for necessity, based on entropy

A vast majority of you used an information theoretic approach (Shannon entropy) to give the necessity proof.

Consider the decision tree of the algorithm, where each internal node is a query of f . There are two possible outcomes for each query, so there are at most two children to each internal node. Each leaf is an outcome, a possible f output by the algorithm. There must be $\geq 2^n$ leaves, so the decision tree has height $\geq n$. But the height of the decision tree is exactly the maximum possible number of queries that the algorithm needs over all cases. So the algorithm will need $\geq n$ queries.

An alternate way to phrase this is using bits of entropy. There are 2^n possible functions f , so it can be represented with $\log_2 2^n = n$ bits of information. Every query to f has two possible outcomes, so it gives 1 bit of information. Therefore we need at least

$$\frac{n}{1} = n$$

queries to even have enough bits to determine f exactly.

(You might be familiar of this approach from e.g. proving that a comparison sorting algorithm requires $\Omega(n \log n)$ comparisons always. The decision tree must have $n!$ leaves, and each internal node only has two children, so the height of the tree is $\geq \log_2(n!) = \Omega(n \log n)$.)

2 Problem 2

Let $f : \mathbb{F}_2^n \rightarrow \{-1, +1\}$ be a Boolean function.

(a) Let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{F}_2^n$ be chosen uniformly at random. Prove that

$$\mathbb{E}[f(\mathbf{x})f(\mathbf{y})f(\mathbf{z})f(\mathbf{x} + \mathbf{y} + \mathbf{z})] = \sum_{S \subseteq [n]} \hat{f}(S)^4.$$

(b) If f is ε -far from every parity function and its negation, prove that

$$\Pr(f(\mathbf{x})f(\mathbf{y})f(\mathbf{z}) \neq f(\mathbf{x} + \mathbf{y} + \mathbf{z})) = \Omega(\varepsilon).$$

2.1 My solution

Part (a). We follow the same approach as in lecture 11 slide 20. We have:

$$\begin{aligned} \mathbb{E}_{\mathbf{x}, \mathbf{y}, \mathbf{z}}[f(\mathbf{x})f(\mathbf{y})f(\mathbf{z})f(\mathbf{x} + \mathbf{y} + \mathbf{z})] &= \mathbb{E}_{\mathbf{x}, \mathbf{y}}[f(\mathbf{x})f(\mathbf{y}) \cdot \mathbb{E}_{\mathbf{z}}[f(\mathbf{z})f(\mathbf{x} + \mathbf{y} + \mathbf{z})]] \\ &= \mathbb{E}_{\mathbf{x}, \mathbf{y}}[f(\mathbf{x})f(\mathbf{y}) \cdot (f * f)(\mathbf{x} + \mathbf{y})] \\ &= \mathbb{E}_{\mathbf{x}}[f(\mathbf{x}) \cdot \mathbb{E}_{\mathbf{y}}[f(\mathbf{y}) \cdot (f * f)(\mathbf{x} + \mathbf{y})]] \\ &= \mathbb{E}_{\mathbf{x}}[f(\mathbf{x}) \cdot (f * (f * f))(\mathbf{x})] \\ &= \sum_{S \subseteq [n]} \hat{f}(S) \cdot \widehat{(f * (f * f))}(S) \\ &= \sum_{S \subseteq [n]} \hat{f}(S)^2 \cdot \widehat{(f * f)}(S) \\ &= \sum_{S \subseteq [n]} \hat{f}(S)^4 \end{aligned}$$

Part (b). Note that (lecture 11 slide 8, apply to these two functions):

$$\mathbb{E}[f(\mathbf{x})f(\mathbf{y})f(\mathbf{z})f(\mathbf{x} + \mathbf{y} + \mathbf{z})] = 1 - 2 \cdot \Pr(f(\mathbf{x})f(\mathbf{y})f(\mathbf{z}) \neq f(\mathbf{x} + \mathbf{y} + \mathbf{z}))$$

Therefore, to give a lower bound to the probability $\Pr(\dots)$, we want to give an upper bound to the expectation $\mathbb{E}[\dots]$. (For sake of convenience, I will use $\dots$ to represent this long expression.)

From part (a), we know

$$\begin{aligned} \mathbb{E}[\dots] &= \sum_S \hat{f}(S)^4 \\ &\leq \sum_S \left(\max_S \hat{f}(S)^2 \cdot \hat{f}(S)^2 \right) \\ &= \max_S \hat{f}(S)^2 \quad \quad \quad (\text{Parseval, } \sum_S \hat{f}(S)^2 = 1) \end{aligned}$$

From lecture 11 slide 9, we have

$$\hat{f}(S) = 1 - 2 \cdot \Pr(f \neq \chi_S).$$

However, we also know that f is ε -far from any parity function or its inverse. Therefore, for any $S \subseteq [n]$,

$$\Pr(f \neq \chi_S) \geq \varepsilon \quad \text{and} \quad \Pr(f = \chi_S) = \Pr(f \neq -\chi_S) \geq \varepsilon \implies \Pr(f \neq \chi_S) \leq 1 - \varepsilon$$

Applying these two inequalities to $\hat{f}(S)$ gives

$$1 - 2(1 - \varepsilon) \leq \hat{f}(S) \leq 1 - 2\varepsilon \implies |\hat{f}(S)| \leq 1 - 2\varepsilon.$$

This is true for all S , so

$$\mathbb{E}[\dots] \leq \max_S \hat{f}(S)^2 \leq (1 - 2\varepsilon)^2.$$

Finally,

$$\begin{aligned} \Pr(f(\mathbf{x})f(\mathbf{y})f(\mathbf{z}) \neq f(\mathbf{x} + \mathbf{y} + \mathbf{z})) &= \frac{1 - \mathbb{E}[\dots]}{2} \\ &\geq \frac{1 - (1 - 2\varepsilon)^2}{2} \\ &= 2\varepsilon(1 - \varepsilon) \end{aligned}$$

Clearly $\varepsilon \leq 1/2$, because f is ε -far from both constant functions $+1$ and -1 . Therefore the probability is lower bounded by ε , proving the claim (with constant 1).

Note: You can improve the constant to be as close to 4 as you want when ε is small enough. The inequality

$$\sum_S \hat{f}(S)^4 \leq \sum_S \left(\max_S \hat{f}(S)^2 \cdot \hat{f}(S)^2 \right)$$

is a bit weak; only one of the terms can reach the maximum. When ε is small enough, you can bound this better using

$$\leq (1 - 2\varepsilon)^4 + (1 - (1 - 2\varepsilon)^2)^2 = 1 - 8\varepsilon + O(\varepsilon^2).$$

(Note that big O notation here hides terms of higher degree, because $\varepsilon \rightarrow 0$.) This leads to the probability being

$$\geq (4 - o(1)) \cdot \varepsilon.$$

2.2 Insight and discussion

The motivation behind this problem is that, we use the BLR linearity test, but with 4 queries instead of 3. As such, the solution also directly follows the proof of BLR linearity test (lecture 11 slides 15–22). I don't think there's anything particularly notable here; you simply work out the same proof.

Why would we use 4 queries instead of 3? Well, 3 queries allows us to determine if f is close or far from the set of parity functions. But for 4 queries, the set is parity functions and their negations. In a way, our intuition says negations of parity functions should be just as "simple" as parity functions themselves. This is not exactly the case (the f is not linear in $\mathbb{F}_2$, just affine), but it is a reason enough.

2.3 Grading criteria and common mistakes

There is no notable error here. In general, I am also more forgiving in grading this problem; I didn't check the full calculations, as long as they look right enough.

2.4 Appendix: (a) Breaking down the expectation

If you don't know convolutions like from the lecture, is there a way to solve part (a)? There is, if you're willing to get dirty.

Recall that (lecture 10 slide 10) a Boolean function can be written in the form

$$f(\mathbf{x}) = \sum_{S \subseteq [n]} \hat{f}(S) \chi_S(\mathbf{x})$$

where

$$\chi_S(\mathbf{x}) = (-1)^{\sum_{i \in S} x_i}$$

is the parity function. Here we have the summation $\sum_{i \in S} x_i$ on the exponent instead of a product $\prod_{i \in S} x_i$ as per the slide, because the domain for $\mathbf{x}$ is $\mathbb{F}_2^n$, not $\{+1, -1\}^n$.

Consider the expression

$$f(\mathbf{x})f(\mathbf{y})f(\mathbf{z})f(\mathbf{x} + \mathbf{y} + \mathbf{z}).$$

Each f is a massive summation over all $S \subseteq [n]$, and there are four f 's here. Any given term can be written in the following form:

$$\hat{f}(X)\chi_X(\mathbf{x}) \cdot \hat{f}(Y)\chi_Y(\mathbf{y}) \cdot \hat{f}(Z)\chi_Z(\mathbf{z}) \cdot \hat{f}(A)\chi_A(\mathbf{x} + \mathbf{y} + \mathbf{z}) \quad (1)$$

where $X, Y, Z, A \subseteq [n]$.

Note that

$$\chi_A(\mathbf{x} + \mathbf{y} + \mathbf{z}) = \chi_A(\mathbf{x})\chi_A(\mathbf{y})\chi_A(\mathbf{z}),$$

so we will rearrange the term on (1) into

$$\hat{f}(X)\hat{f}(Y)\hat{f}(Z)\hat{f}(A) \cdot \chi_X(\mathbf{x})\chi_A(\mathbf{x}) \cdot \chi_Y(\mathbf{y})\chi_A(\mathbf{y}) \cdot \chi_Z(\mathbf{z})\chi_A(\mathbf{z}) \quad (2)$$

Finally, note that

$$\chi_X(\mathbf{x})\chi_A(\mathbf{x}) = \chi_{X \triangle A}(\mathbf{x})$$

where $X \triangle A = (X \setminus A) \cup (A \setminus X)$ is the symmetric difference. Therefore, (2) can further be rewritten into

$$\hat{f}(X)\hat{f}(Y)\hat{f}(Z)\hat{f}(A) \cdot \chi_{X \triangle A}(\mathbf{x}) \cdot \chi_{Y \triangle A}(\mathbf{y}) \cdot \chi_{Z \triangle A}(\mathbf{z}) \quad (3)$$

We will compute the expectation of (3).

Note that $\mathbf{x}, \mathbf{y}, \mathbf{z}$ are all independent from each other, so it is enough to compute $\mathbb{E}[\chi_{X \triangle A}(\mathbf{x})]$ and the same for the other vectors. But we know this expectation: it is 1 if $X \triangle A = \emptyset$ (the vector $\mathbf{x}$ never affects the parity function), and 0 otherwise (exactly half of the vectors will give +1 and the other half will give -1). Therefore, if $X \triangle A \neq \emptyset$, the expectation of (3) is zero.

Similarly, if any of $X \triangle A, Y \triangle A, Z \triangle A$ is nonempty, the expectation is zero. The only remaining case is if all of them are empty, which means all of X, Y, Z, A are the exact same set, say S . In this case, all the parity functions are always 1. Then (3) simply becomes

$$\hat{f}(S)^4 \cdot 1 \cdot 1 \cdot 1.$$

Summing over all S gives the formula.

Note: This method also works for the one in the lecture with 3 queries.

3 Problem 3

Let $f : \{-1, +1\}^n \rightarrow \{-1, +1\}$. If f is monotone, odd, and symmetric, prove that (i) n is odd, and (ii) f is the majority function Maj_n .

3.1 My solution

First, because f is symmetric, we can represent f using another function $g : \{0, 1, 2, \dots, n\} \rightarrow \{-1, +1\}$:

$$f(\mathbf{x}) = g(\# \text{ of arguments in } \mathbf{x} \text{ that are } +1)$$

This is well-defined because f is symmetric. We will focus mainly on g .

Now, f is monotone. This translates to the following monotone property on g :

$$s \leq t \implies g(s) \leq g(t)$$

The proof is straightforward. Let $\mathbf{x}$ be any vector with exactly s arguments being $+1$. Let $\mathbf{y}$ be a vector obtained from $\mathbf{x}$, by toggling some -1 's into $+1$'s so there are t of them. Then by monotonicity of f ,

$$g(s) = f(\mathbf{x}) \leq f(\mathbf{y}) = g(t)$$

Finally, f is odd. This translates to the following property on g :

$$g(n - s) = -g(s)$$

The proof is also straightforward. Let $\mathbf{x}$ be any vector with exactly s arguments being $+1$. Let $-\mathbf{x}$ be its negation, where every component is negated. Then $-\mathbf{x}$ has exactly $n - s$ arguments being $+1$. Then by oddness of f ,

$$g(n - s) = f(-\mathbf{x}) = -f(\mathbf{x}) = -g(s)$$

We now apply the oddness property to the monotone property.

If n is even, then

$$g(n - n/2) = -g(n/2) \implies g(n/2) = 0$$

which is impossible. So if n is even, there is no such function f .

If $n = 2k + 1$ is odd, then

$$g(k + 1) = g(n - k) = -g(k).$$

But we also have $g(k) \leq g(k + 1)$. Since g can only take values -1 and $+1$, we have $g(k) = -1$ and $g(k + 1) = +1$. By the monotonicity property,

$$g(s) = \begin{cases} -1 & \text{if } s \leq k \\ +1 & \text{if } s \geq k + 1 \end{cases}$$

This is exactly the majority function Maj_n .

3.2 Insight and discussion

The symmetric property is extremely powerful, because it lets you reduce f into the function g that only counts how many $+1$'s are in the argument. (This is why transitive-symmetry is a more common property.)

The other properties, like monotone and odd, can be translated into equivalent forms on g . Monotone plus symmetric actually forces g to have a “threshold” form: there exists some threshold c such that, if the input has $\leq c$ arguments equal to $+1$, then g gives -1 , and otherwise g gives $+1$. Then odd forces this threshold c to be exactly the midpoint. (Also, odd plus symmetric by itself forces n odd.)

When preparing this problem, we never intended the two parts (i) and (ii) to be separated, so we do not have separate marks for them. (The draft had “prove that f is the majority function Maj_n for some odd n ”.) Most of you separated the two proofs, but that is fine. This also allows you to notice that (i) only requires f to be odd and symmetric, it doesn't have to be monotone.

3.3 Grading criteria and common mistakes

There is no notable error here, I believe everyone got this correct.

4 Problem 4

Prove that the conductance of the d -dimensional hypercube Q_d is

$$\Phi(Q_d) = \frac{1}{d}.$$

4.1 My solution

To prove this, we need to prove there is a cut that achieves conductance $1/d$, and that all cuts have conductance $\geq 1/d$.

4.1.1 The conductance is $\leq 1/d$

Take one half of the hypercube $Q_d = \{-1, +1\}^d$ as our S . By one half, I mean a half obtained by splitting with a hyperplane. For instance, we can take S to be the set of all vertices with first coordinate $+1$.

Note that each side has 2^{d-1} vertices, so

$$\text{vol}(S) = d \cdot 2^{d-1}.$$

In addition, $V \setminus S$ also has 2^{d-1} vertices, so the two parts are equally large; we may take S as the smaller part.

Also, we claim

$$|E(S, V \setminus S)| = 2^{d-1}.$$

To see this, simply note that every vertex in S has exactly one edge outgoing S , namely the edge in the same axis as the fixed coordinate. In our example, our S has first coordinate $+1$, so the edges in the cut are those that change the first coordinate, e.g. $(+1, +1, +1, \dots, +1)$ to $(-1, +1, +1, \dots, +1)$.

Therefore,

$$\Phi(Q_d) \leq \Phi(S) = \frac{|E(S, V \setminus S)|}{\text{vol}(S)} = \frac{2^{d-1}}{d \cdot 2^{d-1}} = \frac{1}{d}.$$

4.1.2 The conductance is $\geq 1/d$

To give a lower bound on the conductance, we will represent the cut as a Boolean function. Take any $S \subseteq Q_d$, and construct a Boolean function $f : \{-1, +1\}^d \rightarrow \{-1, +1\}$ as follows: $f(v) = +1$ if $v \in S$ and -1 otherwise.

Let $I[f]$ be the total influence of f (lecture 12 slide 23). Recall that in Q_d , we have $|E(Q_d)| = d \cdot 2^{d-1}$. So

$$\frac{I[f]}{d} = \frac{|E(S, V \setminus S)|}{|E(Q_d)|} \implies |E(S, V \setminus S)| = I[f] \cdot 2^{d-1}$$

Now we consider the conductance:

$$\Phi(S) = \frac{|E(S, V \setminus S)|}{\min\{\text{vol}(S), \text{vol}(V \setminus S)\}} = \frac{I[f] \cdot 2^{d-1}}{d \cdot \min\{|S|, |V \setminus S|\}}$$

We will now look into $I[f]$. By Poincaré's inequality, $I[f] \geq \text{Var}[f]$. By definition of variance, we have $\text{Var}[f] = \mathbb{E}[f^2] - \mathbb{E}[f]^2$. Because f has codomain $\{-1, +1\}$, we have $f^2 = 1$ always, so $\mathbb{E}[f^2] = 1$. Combining these together,

$$I[f] \geq 1 - \mathbb{E}[f]^2 = (1 + \mathbb{E}[f])(1 - \mathbb{E}[f])$$

However, we know

$$\mathbb{E}[f] = \frac{|S|}{2^d} \cdot (+1) + \frac{2^d - |S|}{2^d} \cdot (-1) = \frac{|S| - 2^{d-1}}{2^{d-1}}$$

Therefore,

$$\begin{aligned}
 I[f] &\geq (1 + \mathbb{E}[f])(1 - \mathbb{E}[f]) \\
 &= \frac{|S|}{2^{d-1}} \cdot \frac{2^d - |S|}{2^{d-1}} \\
 &= 2^{-2(d-1)} \cdot |S| \cdot |V \setminus S| \\
 &\geq 2^{-2(d-1)} \cdot \min\{|S|, |V \setminus S|\} \cdot 2^{d-1} \\
 &= 2^{-(d-1)} \cdot \min\{|S|, |V \setminus S|\}
 \end{aligned}$$

Finally, we simply substitute this to the conductance:

$$\Phi(S) = \frac{I[f] \cdot 2^{d-1}}{d \cdot \min\{|S|, |V \setminus S|\}} = \frac{2^{d-1} \cdot (2^{-(d-1)} \cdot \min\{|S|, |V \setminus S|\})}{d \cdot \min\{|S|, |V \setminus S|\}} \geq \frac{1}{d}$$

This is true for every cut $(S, V \setminus S)$. Therefore

$$\Phi(Q_d) = \min_S \Phi(S) \geq \frac{1}{d}.$$

4.2 Insight and discussion

The bulk of the problem is to show the conductance is $\geq 1/d$. I think the argument is incredibly pretty; it links up various results about Boolean functions that you've seen. First, $I[f]$ is related to the size of the cut, directly influencing conductance. Second, we have Poincaré's inequality to give $I[f] \geq \text{Var}[f] = \mathbb{E}[f^2] - \mathbb{E}[f]^2$. Third, we have $f^2 = 1$ always by definition – although you probably think of Parseval's theorem as well, as the same idea is used there. Fourth, $\mathbb{E}[f]$ is related to $|S|$ directly, so this helps to offset the $\min\{\text{vol}(S), \text{vol}(V \setminus S)\}$ term in the conductance. They all just link up so perfectly.

That said, you also need to show the conductance is $\leq 1/d$. You can find one such cut pretty intuitively; cutting a hypercube in half is a natural operation, and it directly gives you the desired conductance. Or else, you can also investigate your proof. You need equalities everywhere: $I[f] = \text{Var}[f]$ which means f is linear (each term only depends on one variable), and $\max\{\text{vol}(S), \text{vol}(V \setminus S)\} = 2^{d-1}$ which means S has exactly half of the vertices. This immediately gives rise to the example.

4.3 Grading criteria and common mistakes

There is no notable error here. It is important to show that the conductance is $\leq 1/d$ by e.g. showing a cut that achieves it, but I believe all of you did so.

4.4 Appendix: An inductive proof without using Boolean functions

A student submitted an inductive proof to prove $\Phi(Q_d) \geq 1/d$, which does not use Boolean functions. On one hand, we put this problem here so you can make use of various theorems about Boolean functions. On the other, I suppose it is not surprising that you can solve it without Boolean functions, given the problem statement has nothing to do with them too.

We will show, for any $S \subseteq V(Q_d)$, that

$$|E(S, V(Q_d) \setminus S)| \geq \min\{|S|, |V(Q_d) \setminus S|\}$$

by induction on d . In the proof, we will assume $|S| \leq |V(Q_d)|/2 = 2^{d-1}$.

The base case is $d = 0$, where Q_d is just a single isolated vertex. Clearly the inequality is satisfied, because $|S| = 0$.

Now consider a cut S of Q_d for $d \geq 1$. Divide the hypercube into two halves L and R (using a hyperplane), and let $S_L = S \cap L$ and $S_R = S \cap R$. Without loss of generality, assume $|S_L| \leq |S_R|$. We have the following:

$$\begin{aligned} |E(S, V(Q_d) \setminus S)| &= |E(S_L, L \setminus S_L)| && \text{(same side)} \\ &+ |E(S_L, R \setminus S_R)| && \text{(across sides)} \\ &+ |E(S_R, L \setminus S_L)| && \text{(across sides)} \\ &+ |E(S_R, R \setminus S_R)| && \text{(same side)} \end{aligned}$$

As broken down in the above, each term can be designated as a cut on the same side (in L or in R), or a cut that is across sides.

For a cut on the same side, we use the inductive hypothesis. We have

$$|E(S_L, L \setminus S_L)| \geq \min\{|S_L|, |L \setminus S_L|\} \quad \text{and} \quad |E(S_R, R \setminus S_R)| \geq \min\{|S_R|, |R \setminus S_R|\}$$

Furthermore, because we assumed $|S_L| \leq |S_R|$ and $|S| = |S_L| + |S_R| \leq 2^{d-1}$, it follows that $|S_L| \leq 2^{d-2}$, so it is the smaller side in the cut in L . Therefore

$$|E(S_L, L \setminus S_L)| \geq |S_L| \quad \text{and} \quad |E(S_R, R \setminus S_R)| \geq \min\{|S_R|, |R \setminus S_R|\}$$

For the two terms across sides, we compute their sum together as follows. Let v be a vertex in Q_{d-1} , and let v_L, v_R be the corresponding vertices on Q_d on the L and R sides respectively. Then $v_L v_R$ is an edge in the cut iff exactly one of them is in S , and in this case, it will be counted by one of the two terms.

However, when summing over all v , there are at least $|S_R| - |S_L|$ such vertices v whose cross edge is in the cut. This is because $|S_R| \geq |S_L|$, so at least $|S_R| - |S_L|$ of the vertices in S_R do not have their corresponding partners in S_L . Each such vertex is some v to be counted.

Therefore

$$|E(S_L, R \setminus S_R)| + |E(S_R, L \setminus S_L)| \geq |S_R| - |S_L|.$$

Finally, we add up all the inequalities:

$$\begin{aligned} |E(S, V(Q_d) \setminus S)| &\geq |S_L| && \text{(inside } L) \\ &+ (|S_R| - |S_L|) && \text{(across sides)} \\ &+ \min\{|S_R|, |R \setminus S_R|\} && \text{(inside } R) \end{aligned}$$

We finally divide into two cases.

If S_R is the smaller side in R , then

$$|E(S, V(Q_d) \setminus S)| \geq |S_R| + |S_R| \geq |S_L| + |S_R| = |S|.$$

If S_R is the larger side in R , then

$$|E(S, V(Q_d) \setminus S)| \geq |S_R| + |R \setminus S_R| = 2^{d-1} \geq |S|.$$

Thus we have proven the claim by induction.

Finally, when $|S| \leq |V(Q_d)|/2$, note that $\text{vol}(S) = d \cdot |S|$, so

$$\Phi(S) = \frac{|E(S, V(Q_d) \setminus S)|}{\text{vol}(S)} \geq \frac{|S|}{d \cdot |S|} = \frac{1}{d}.$$

This holds for all S , so $\Phi(Q_d) \geq 1/d$.